



LAB 05

PH142 Summer 2026

Announcements

- 7/22: Quiz12, Lab05
- 7/23: Quiz13
- 7/24: Quiz14

Data Project Part II is due next Friday. Start working on it early!



Lab05 Lecture Review

Independence of Events

Independent Events

- A and B are independent if knowing whether A occurs does not change the probability of B

Dependent Events

- A and B are dependent if knowing whether A occurs does change the probability of B

Lab05 Lecture Review

Screening Tests: Key Terms

Sensitivity: Probability the test is positive given disease present

- Sensitivity = $P(\text{Test+} \mid \text{Disease+})$

Specificity: Probability the test is negative given disease absent

- Specificity = $P(\text{Test-} \mid \text{Disease-})$

Sensitivity and specificity are properties of the test. They do not change based on prevalence of disease in the population.

Lab05 Lecture Review

Predictive Values

Positive Predictive Value (PPV): Probability the disease is present given a positive test result

- $PPV = P(\text{Disease+} \mid \text{Test+})$

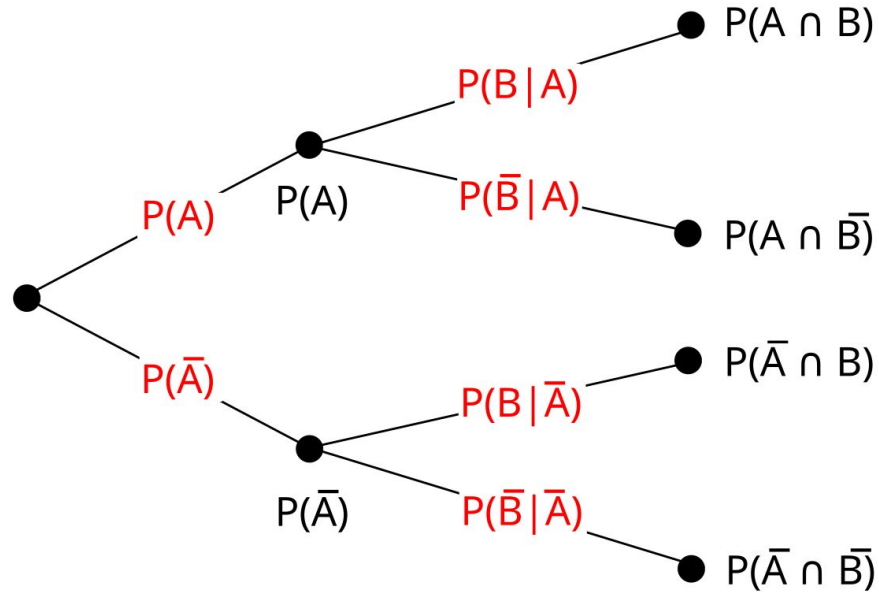
Negative Predictive Value (NPV): Probability the disease is absent given a negative test result

- $NPV = P(\text{Disease-} \mid \text{Test-})$

PPV and NPV change with the prevalence of disease in the population.

Lab05 Lecture Review

Tree Diagrams



Lab05 Lecture Review

The Normal Distribution

The normal distribution is a bell-shaped, symmetric distribution, with notation $\mathbf{X} \sim \mathbf{N}(\mu, \sigma)$

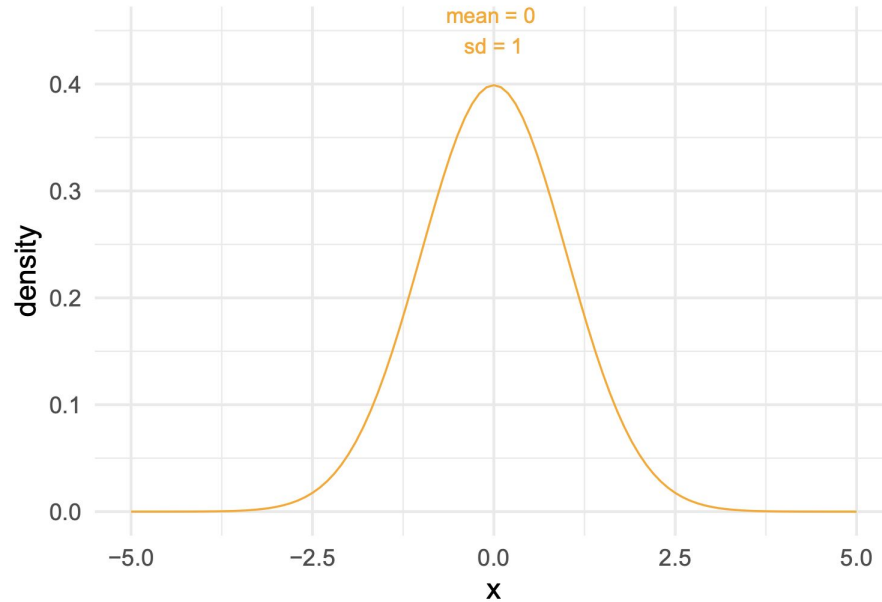
- μ = mean
- σ = standard deviation

R Functions:

- `pnorm()` → outputs the probability of value x or below
- `rnorm()` → generates random draws from the distribution

Lab05 Lecture Review

The Normal Distribution



Lab05 Lecture Review

Z-Scores

Z-score formula: $z = \frac{x - \mu}{\sigma}$

x = data point

μ = mean

σ = standard deviation

The Z-score tells you how many standard deviations the data point is from the mean.



LAB 05 Walkthrough

Lab Submission

- Follow the directions on the LAB05 file
- Submit using the **Terminal Tab** (next to the console in the bottom left pane)
 - Copy and paste the given line into the terminal
 - Follow prompts (NOTE: the terminal will **not** show your password being typed out!)
- **CHECK IN GRADESCOPE THAT ALL YOUR TESTS PASSED**